LipLearner: Customizing Silent Speech Commands from Voice Input using One-shot Lipreading

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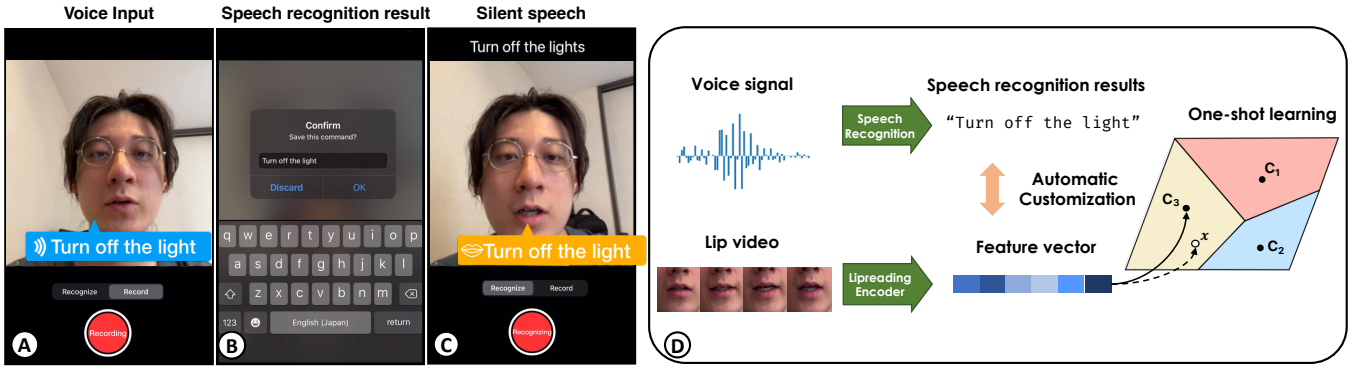

Figure 1: In-situ silent speech command customization using LipLearner. (A) To add a new command, the user says the command aloud while holding down the recording button. (B) The speech recognizer translates the user's voice into text. The user can edit the result before confirming. (C) The command is then available for silent speech. (D) LipLearner automatically learns lip movements and the corresponding speech content during voice input.

ABSTRACT

We present LipLearner, a lipreading-based silent speech interface that enables in-situ command customization on mobile devices. By leveraging contrastive learning to learn efficient representations from existing datasets, it performs instant fine-tuning for unseen users and words using one-shot learning. To further minimize the labor of command registration, we incorporate speech recognition to automatically learn new commands from voice input. Conventional lipreading systems provide limited pre-defined commands due to the time cost and user burden of data collection. In contrast, our technique provides expressive silent speech interaction with minimal data requirements. We conducted a pilot experiment to investigate the real-time performance of LipLearner, and the result demonstrates that an average accuracy of 91.33% is achievable with only one training sample for each command.

CCS CONCEPTS

• **Human-centered computing → Interaction techniques**.

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KEYWORDS

Silent Speech Interface, Lipreading, One-shot Learning, Customization

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1 INTRODUCTION

To tackle the privacy and social acceptance problems in voice user interfaces, Silent Speech Interface (SSI) has emerged as a promising alternative that exploits non-acoustic signals to enable speech recognition without voice. SSI provides seamless and confidential interactions in various situations, especially in those where voice interaction is inappropriate or unavailable. Recent research on SSI has proposed to use various sensing methods such as Electromyography (EMG) [3, 8], ultrasound imaging [5], capacitive sensing [6] and video camera (lipreading) [4, 9, 10] to track the movement of speech articulators and decode silent speech.

Among all approaches, lipreading has minimal device requirements while providing rich information with high temporal and spatial resolution. The proliferation of smartphones, which are mostly equipped with a high-quality front camera, makes it even easier to access. However, building expressive yet robust lipreading systems remains challenging because training the neural network

requires a large amount of data. In previous research, the user has to provide hundreds of samples (typically 20∼30 samples per command [4, 10]) before they can use the system. As a result, whether setting up the system or involving novel commands could lead to huge physical and mental burden, and it is also time-consuming to update the model every time the command set changes. Therefore, existing methods only provide only a limited number of pre-defined commands, and the rich interaction space in silent speech is waiting to be mined.

To liberate the expensiveness of lipreading as well as minimize the user burden in the data collection process, we propose LipLearner to leverage contrastive learning to exploit knowledge from existing datasets, allowing instant fine-tuning for unseen users and words using one-shot learning. Thus, the user can freely define any phrases as their own commands by providing only one sample for each command. Notably, although our model is pre-trained on an English corpus, it is trivial to enable non-English language or even non-verbal lip gestures for silent speech input. We further introduce a multimodal command registration technique, which automatically learns silent speech from voiced speech by remembering lip movements and the speech content recognized from voice signal. This process can be invisible to the user by learning from daily voice interactions, offering a further seamless experience of tailored silent speech interaction.

2 LIPLEARNER

2.1 Dataset and Preprocessing

We use a large-scale lipreading dataset LRW [1] to pre-train the feature extractor for one-shot lipreading. The LRW dataset consists of video segments extracted from the broadcast content of BBC news with up to 1000 utterances of 500 different words spoken by hundreds of speakers. The dataset also covers different recording conditions such as lighting, background, and camera perspective, which fits well to the varying environments where people use mobile devices in everyday life. Therefore, we expect LRW can maximize the model's capability for in-the-wild recognition.

2.2 Model Architecture

We adopt a Temporal Convolutional Network (TCN) based on the architecture proposed in [2] as the backbone of our encoder model. The front end is a ResNet-18 network modified for video input with a 3D convolution kernel (size $5 \times 7 \times 7$). After a global pooling layer, the output is reshaped into $T \times 512$ (T denotes time). The back end uses a 3-Layers bidirectional Gated Linear Unit (GRU) followed by an average pooling and a fully connected layer. The model takes an input of 29 frames center-cropped to 88×88 pixels and outputs a 500-dimensional vector.

2.3 Contrastive Learning

The goal of contrastive learning is to learn an embedding space in which similar samples are close to each other while dissimilar ones are far apart. We perform a semi-supervised version of contrastive learning (Fig 2) to train a model that can easily generalize to unseen users and commands by combining with one-shot learning. First, we randomly select one sample from each of N classes as group A, and then select another group B from the same

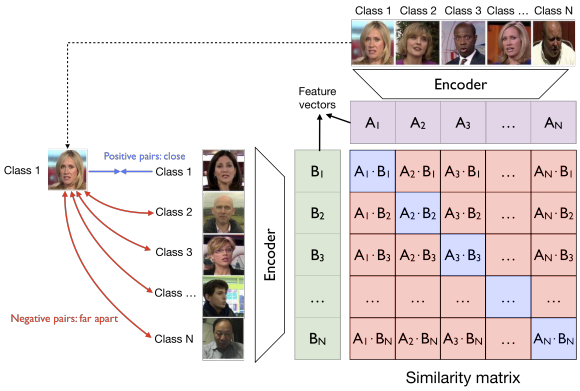

Figure 2: Contrastive learning on LRW dataset. Illustrated with the first class in the batch as an example but applies to all classes.

N classes without replacement. So both group A and B consist of N samples that are all from different classes. Next, a similarity matrix is calculated among the embedding vectors extracted by the encoder model. Here, the cosine similarity is measured by the scalar product of two vectors $S_(i, j) = \boldsymbol{A_i} \cdot \boldsymbol{B_j}$, where $i, j \in [0, N]$ denote the class index. The model contrastively learns from the N positive pair examples from the same classes and the $N^2 - N$ negative pairs from different classes using the InfoNCE loss [11] $\mathcal{L} = -\frac{1}{2N}(\sum_{i=1}^{N} \log \frac{e^{S_{i,i}}}{\sum_j e^{S_{i,j}}} + \sum_{j=1}^{N} \log \frac{e^{S_{j,j}}}{\sum_i e^{S_{i,j}}})$. The objective is to maximize the similarity between positive pairs (the diagonal values in the matrix) and vice versa. We trained our model for 120 epochs with an initial learning rate of 3×10^{-4} delayed by a cosine annealing scheduler.

2.4 In-situ Customization and Evaluation

To enable in-situ silent speech customization, we integrate the pre-trained lipreading encoder with one-shot learning to transfer to unseen commands and users. As shown in Fig 1A, we built a prototype of LipLearner on iPhone 13 Pro using a multimodal command registration technique that allows the user to add a new command using voice input. During the utterance, the user keeps holding down the record button to record the lip movements using the front camera and the voice using the microphone. The video is streamed to a laptop PC with an RTX 3080 GPU for recognition. First, the frames are cropped into square regions of interest (ROI) with the mouth centered using the MediaPipe face landmark detector [7]. The ROI is converted to a grayscale image and then resized to 88 (H) × 88 (W) pixels and then encoded into feature vectors. For each user, we train an SVM classifier on their own data to maximize the accuracy (training time is approximately 0.1 second). The SVM learns the decision boundary in a one-shot manner where only one sample is used for each command (Fig 1D). Meanwhile, the voice signal is translated into texts by the built-in speech recognizer [1] on iOS 15.6. The user can edit or correct the text if there is an error. With the text as the label, the model can recognize silent speech input into commands (Fig 1C).

[1] SFSpeechRecognizer | Apple Developer Documentation https://developer.apple.com/documentation/speech/sfspeechrecognizer

Using this proof-of-concept prototype, we conducted a pilot experiment with 3 participants (2 male, 1 female), which took around 10 minutes. In the registration session, the participant was asked to *read aloud* the 10 sentences from the OuluVS corpus [12] as commands in random order. In the test session, the participant issued each command 5 times in a *silent* manner. Those commands are recognized in real time and the results are displayed at the top of screen. Overall, 10 sentences $\times$ 5 repetitions $\times$ 3 participants = 150 trial were completed, achieving a promising average accuracy of 91.33% $\pm$ 2.49%.

3 CONCLUSION AND FUTURE WORK

We presented LipLearner, a one-shot lipreading system that enables silent speech commands with handy initialization and in-situ customization. LipLearner liberates users from the costly and time-consuming loop of data collection and model re-training, thereby opening the way for shaping creative and tailored silent speech interactions. In the future, we plan to conduct a comprehensive study to evaluate our system against different environmental factors, such as illumination and view angle, to verify its feasibility in real-world settings.

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